

packages/graph-core/src/analyzer/layer-detector.ts

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Overview

A new file `packages/graph-core/src/analyzer/layer-detector.ts` adds lightweight layer detection and LLM-based inference utilities.

The module imports core types (`KnowledgeGraph` , `Layer`) from `../types.js` (line 1) and exports several helpers.

Key changes

- **Interface** `LLMLayerResponse` (lines 6-10) defines the LLM-returned layer shape.
- **Pattern table** `LAYER_PATTERNS` (lines 16-67) lists directory-pattern → layer mappings.
- **Helper** `toLayerId` (lines 72-74) normalises a layer name to a kebab-case ID.
- **File-to-layer matcher** `matchFileToLayer` (lines 80-97) checks path segments against `LAYER_PATTERNS` .
- **Heuristic detector** `detectLayers` (lines 105-144) walks the graph, assigns file nodes to layers via `matchFileToLayer` , and creates a "Core" layer for unmatched files.
- **Prompt builder** `buildLayerDetectionPrompt` (lines 150-169) lists all file paths for an LLM prompt.
- **Response parser** `parseLayerDetectionResponse` (lines 177-217) extracts JSON from raw or fenced responses, validates entries, and returns an array of `LLMLayerResponse` .
- **LLM application** `applyLLMLayers` (lines 227-284) maps file nodes to LLM-defined layers using `filePatterns` ; unmatched files go to an "Other" layer.

Impact

- Adds deterministic layer assignment logic and LLM integration without altering existing APIs.
- All operations run in linear time over graph nodes; no new I/O or recursion.
- Centralises layer logic in a single module, simplifying reuse across services.

Risks & follow-ups

- `matchFileToLayer` may misclassify files when directory names overlap; test against edge-case paths.
- `parseLayerDetectionResponse` assumes a JSON array; validate with varied LLM outputs (code fences, plain text).
- `applyLLMLayers` lazily creates an "Other" layer; ensure consumers handle missing layers.
- Path normalization replaces backslashes but does not resolve relative segments (`..`); run integration tests on mixed OS environments.